A010 – Purpose, Structure, and Reading Guide of the A-Series

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Contents

.1 Use of AI Tools

This document series was created with the assistance of AI tools. The notice appears at the front, not in small print, and belongs as its own section in every document of the series.

The division of labour is clear. The questions, the theoretical stipulations, the geometric ideas, and all substantive decisions come from the author. The AI tools are tools in the literal sense: they formulate, compute, check for contradiction, create verification scripts, find inconsistencies between documents, and translate between representations. They make no substantive decisions and generate no theses.

This separation is not merely an attribution but is methodologically essential. A language model generates plausible-sounding text even where there is no substance – exactly the type of error that the layer marking of the following section is designed to counter. Therefore, throughout this series: every computational statement has a verification script that can be independently checked; every number has a tolerance standard; every derivation step has a layer marker. What does not pass this test does not appear as a result in the text.

Responsibility for content and errors rests entirely with the author.

.2 What Is Not Part of the A-Series

Two areas are explicitly excluded.

The cosmological sector. The A-series does not include it. The corresponding documents from the legacy corpus remain and retain their numbers, but are not carried into the new structure. The reason is a difference in standards: the geometric core derives results from ξ; the cosmological sector works throughout with imported scales and declared calibrations. Mixing both in one series would obscure precisely the separation that motivates the new structure.

Predictions beyond the fundamental quantities. The only quantities carried as predictions are those that follow directly from the apparatus and at which the theory can fail: the tau mass, the ξ-order of the CHSH deviation, the fractal dimension D_f, and the fundamental length L_0. Everything else that appeared as a prediction in earlier documents appears here as a structural statement, or not at all.

.3 Why a New Series

The existing FFGFT document series grew historically. Its numbers follow the order of creation, not the subject matter. Over roughly three years, more than two hundred German-language documents accumulated – individually comprehensible, but collectively exhibiting four properties that harm the whole.

First, reference decay. A substantial portion of the early documents is rarely cited by the later ones. The content has not become wrong, but it is presented more completely and precisely elsewhere. The old number persists and continues to be found.

Second, corrections sit outside the texts. A separate correction register amends documents from the outside instead of modifying them; it records corrections, refinements, and revisions as dated entries. A reader of a single old document therefore reads the uncorrected state. The working rule of not modifying older documents was methodologically correct – it keeps the development history honest – but it shifts the burden onto the reader.

Third, multiple coverage. The mass sector appears in at least five documents, the concept of time in seven, the information question in six. Each of these presentations has its occasion, but there is no single place that says: *this* is the current state.

Fourth, layers are typographically invisible. A statement derived from ξ, an algebraically proved translation, and a plausibility sketch appear in the same typeface side by side. This is the most dangerous of the four properties, because it inflates the theory's apparent claim.

[STIPULATION] The A-series addresses all four points with the same measure: subject order instead of chronological order, corrections integrated instead of appended, one topic in one place, layer status declared per statement.

.4 Introductory Reading Before the A-Series

A reader entering the A-series encounters stipulations, layer markers, and geometric structures from A020 onward. Two documents in the legacy corpus prepare for this – in different ways.

"The Journey" (legacy corpus, popular science) describes the T0 theory in everyday language – with analogies, without formulas, without requiring a physics background. A reader who reads it before A010 arrives with a picture in mind: why a single number ξ might suffice, what time-mass duality means, what is different about this approach from established theories.

"FFGFT in Plain Language" (legacy corpus, narrative-technical) is the narrative introduction to the Lagrangian core. It translates the central Lagrangian formulations into accessible language – using images from acoustics (vibrating string, resonator, Pythagorean comma) that appear in FFGFT itself as structural parallels, not arbitrary metaphors. Every statement is anchored in the formal documents. A reader who wants to understand how the theory computes before seeing the computations should read this document.

Both documents are not part of the A-series and are not superseded by it. They are independent and remain so.

.5 What the A-Series Is Not

[STIPULATION] It is *not* a revision in the sense of reappraisal. Nothing is retracted that has not already been retracted in the correction register. The content is the same; the order is new.

It does not physically replace the old documents. The legacy corpus is fully preserved, unchanged, under its old numbers. It is the development history and remains citable. The A-series is the version to read when one wants to know what the theory claims – not how it got there.

And it is not an abridgment. Where an old text is more detailed than the corresponding A-section, the latter refers to it. What is to be lost is redundancy, not substance.

.6 The Three Layers

FFGFT has always had three methodological layers. What is new is that they are marked in the A-series – per statement, not per document.

[K] Core. Derived from the geometry and the single parameter ξ. Example: the Koide relation $Q = 2/3$ as operator condition $r = \sqrt{2}$; the non-closure of the recursion; $D_f = 3 - \xi$.

[B] Bridge. Algebraically proved translation between two representations. Example: the bijection $\mathscr{H} = L^2(T^4) \otimes \mathbb{C}^3$; the DFT diagonalisation of the Z_3 circulant. A bridge carries everything established on one side, and proves nothing that was not already there.

[S] Sketch. Plausibility, analogy, candidate. Example: the neutrino sector; $N_0 \approx 38.6$; the fine value of the refraction strength δ.

A fourth marker is added that is not a layer but a booking category:

[STIPULATION] Declared, not derived. Example: $\tilde{T} \cdot m = 1$; the choice of T^4; the numerical value $\xi = 4/30000$; the reference point m_μ/m_e.

[B] The purpose of the markers is not modesty but testability. A theory that names its stipulations can be measured by how few it needs. A theory that conceals them cannot.

[STIPULATION] The markers [K] and [B] are internal claim-state markers, not external certifications. [K] states that a claim follows within the declared core; [B] states that a bridge is algebraically established under its stated conditions. Neither marker by itself establishes empirical truth, external physical validity, or a unique ontology.

.7 The Tolerance Standard

Every numerical statement in the A-series names the standard against which it is tested. This is a consequence of an experience older than the theory: a measuring instrument is always less capable than the reality it measures. Exact agreement on the last digit is therefore neither achievable nor the goal.

[STIPULATION] Completeness in the A-series means: *no residual remains that exceeds the tolerances and can only be closed by a free parameter.* A residual within the determination tolerance is not a deficiency but the normal case. A residual that exceeds the tolerance is a statement – and is booked as such, not rounded away.

In practice: at every deviation, three quantities appear together – the value, the measurement precision of the comparison quantity, and the determination tolerance of the theoretical quantity. Only the ratio of these three decides whether a deviation is meaningful.

.8 Citation Policy

[STIPULATION] The A-series carries no reference lists. The rule:

Standard physics without source. Textbook knowledge – constants, standard formulas, established results (Dirac equation, Casimir formula, CHSH bound, Koide relation, Landauer principle) – is used without citation. Name references (Bell, Tsirelson, Weyl) are concept labels, not literature references.

Specific external measurements with inline source. Where a concrete, dated measurement by a third party with error bars is claimed, the source appears in parentheses directly at the number.

Own measurements with reference to the measurement protocol. Hardware runs and own measurement campaigns refer to raw data and evaluation scripts in the corpus, so that the claim is independently verifiable.

.9 Structure and Numbering

The series has four blocks.

Block 0 (A010–A090): Foundation. Stipulations, geometry, fractal correction, recursion, time, Hilbert space, field theory, units, projection chain. A reader who has worked through this block knows the apparatus completely – without a single result. Current documents: A010, A015, A020, A030, A040, A050, A060, A070, A075, A080, A085, A090.

Block 1 (A100–A190): Sectors. Leptons, Koide relation, $\theta = 2/9$, fine-structure constant, g-2, gravitation, gravitational constant, quarks and neutrinos, Bell and hardware, information, Standard Model. The results appear here, each referred back to the apparatus from Block 0. Current documents: A100, A105, A110, A120, A130, A135, A138, A140, A142, A145, A150, A160, A165, A180, A190.

Block 2 (A200–A250): Methodology and assessment. The distinction of law, norm, and stipulation; the test quantities; the tolerance standard; the open questions; the demarcation from external frameworks; the index. Current documents: A200, A210, A220, A230, A240, A250.

Block 3 (A260–): Extensions. Topics without back-reference from earlier blocks – self-contained units that supplement the core but are not embedded in the main flow. Current documents: A260 (Casimir effect), A261 (scale hierarchy), A262 (c-convention and $E = m$), A263 (Dirac complete), A264 (asymmetry and CP), A265 (spectral shift and SI anchor), A266 (unit test and SI back-computation), A267 ($\alpha = 1$ as stipulation).

Methodological status of Block 3. A260–A264 and A265 are physics result documents without back-reference from Block 0/1. A266 (unit test) and A267 ($\alpha = 1$ convention) are convention and methodology documents – they contain computational tools, not new theoretical results. This does not change their place in Block 3, but the reader should know that the claim differs.

[STIPULATION] Numbers within blocks increase in steps of ten. Intermediate numbers (A015, A075, A138, A142, etc.) are assigned when a later document refers back to an earlier one and the ordering requires it. A document without back-reference from the existing block goes to the end – in Block 3, not as an intermediate number. That is the rule guiding the decision: not the topic, but the reference.

.10 Sources

Every A-document names in its closing section the old numbers whose content it subsumes. This is not a reference list but a replacement declaration: a reader who has read the A-document need not read the listed old documents to know the current state.

The reverse direction – from each old number to its A-successor – is collected in A250. There it is also recorded which entries of the correction register are incorporated where, and which old documents deliberately have no successor because their content is outdated or purely historical.

.11 Language Status

The A-series is initially produced in German only. This is a deliberate departure from the previous rule of keeping every change bilingual. The reason is practical: the structure must settle first. An English version is produced when a block is complete and stable – then in one pass from the finished German state, not change by change.

Until then, the bilingual legacy series remains the reference for external correspondence.

.12 Symbol Glossary of the A-Series

The following table explains the symbols used consistently throughout the entire A-series. Symbols that appear only in a single document are introduced there at their first occurrence.

Symbol	Meaning
Fundamental constants and parameters	
$\xi = 4/30000$	Geometric fundamental constant of FFGFT; $\xi = 4/3 \times 10^{-4}$
ℓ_P	Planck length, $\ell_P = \sqrt{\hbar G/c^3} \approx 1.616 \times 10^{-35}$ m
$\hbar$	Reduced Planck constant
c	Speed of light (in natural units: $c = 1$)
α	Fine-structure constant, $\alpha \approx 1/137.036$
φ	Golden ratio, $\varphi = (1+\sqrt{5})/2 \approx 1.618$
FFGFT-specific quantities	
$D_f = 3 - \xi$	Fractal dimension of the FFGFT torus
$L_0 = \xi \cdot \ell_P$	Minimal length scale (sub-Planck), $L_0 \approx 2.15 \times 10^{-39}$ m
$L_T = \ell_P/\xi$	Torus fundamental period, $L_T \approx 7500\,\ell_P$
$\tau_0 = \xi \cdot t_P$	Minimal time scale; by $c = 1$ the same quantity as L_0 in time dress (four-dress principle, A080)
$K_{\text{frak}} = 1 - 100\xi$	Fractal correction factor, $K_{\text{frak}} \approx 0.9867$
$E_0 = \sqrt{m_e m_\mu}$	Characteristic energy, $E_0 \approx 7.398$ MeV
$\tilde{T} \cdot m = 1$	Time-mass duality (stipulation); $\tilde{T}$ is the intrinsic time field
$v = 246$ GeV	Higgs vacuum expectation value (energy unit)
Layer markers	
[STIPULATION]	Axiom of the theory; not further derived
[B]	Bridge; algebraically proved
[K]	Core; numerically verified derivation
[S]	Sketch; plausible but not fully worked out
Mathematical structures	
$T^4/\mathbb{Z}_3$	Compact 4-torus with $\mathbb{Z}_3$ orbifold structure
$\mathcal{H} = L^2(T^4) \otimes \mathbb{C}^3$	Hilbert space of the theory
$\pi_1(T^4) \cong \mathbb{Z}^4$	First homotopy group of the 4-torus
$P_i = \lvert\mathbf{e}_i\rangle\langle\mathbf{e}_i\rvert$	Projector onto mode i
$\Phi = \sum_i m_i P_i$	Mode operator (field operator)
Masses and ratios	
m_e, m_μ, m_τ	Electron, muon, tau mass
$m_i = r_i \cdot \xi^{p_i} \cdot v$	FFGFT mass formula; $r_i \in \mathbb{Q}$ rational prefactor, $p_i \in \mathbb{Q}$ exponent
(n, l, j)	Quantum numbers: principal, angular momentum, spin

.13 Independent Documents of the Legacy Corpus

[STIPULATION] The A-series subsumes 102 documents of the legacy corpus. The remainder – more than 200 numbers – stands independently. This section lists the main thematic groups of these independent documents, so the reader knows where the legacy corpus ends and where the A-series begins. Complete mapping: A250.

Cosmological sector (excluded, Dok. 036, 063, 064, 140 et al.). Hubble constant, cosmic microwave background, vacuum field at galactic scales, Λ_{cosmo}. This sector is not wrong – it is of a different ambition: it works throughout with imported scales and calibrations that the A-series cannot reproduce. Excluded, not superseded.

Gauge sector and QFT extensions (Dok. 186, 192, 281 et al.). Complete Feynman rules, renormalisation group beyond ξ-corrections, Yang-Mills formulation. These documents treat topics declared as open points in the A-series (A190, A240). They remain independent because the projection mapping is still missing.

IPI framework comparisons (Dok. 246, 251, 252, 265, 267, 269, 271, 272, 274, 283, 285, 294, 297, 299 et al.). Structural comparisons with HLV (Helix-Light-Vortex), RA/PMT, Dot Theory, SICC, and other works from the IPI community. These documents are explicitly not FFGFT statements but comparison documents: they investigate where external frameworks and FFGFT share points of contact. The A-series takes over none of these comparisons (A240 names the contact points without carrying them).

Quantum hardware and photonic systems (Dok. 083–085, 161, 167, 173–176, 183 et al.). Photonic chips in communications technology, quantum dots, Shor's algorithm on photonic platforms, Google Willow analysis, Ising machines. These documents apply FFGFT ideas to concrete hardware systems. The application is independent and not back-anchored in the A-series core.

Popular and narrative introductions (Dok. 002, 004, 185, 191 et al.). "The Journey" (narrative introduction without formulas), FFGFT for communications engineers, video scripts. These documents are deliberately not part of the A-series – they are the access point for readers without physics prerequisites and remain so. They are not superseded; they are complementary.

AI and consciousness (Dok. 100, 207 et al.). T0 theory and consciousness; consciousness and bridges. These documents treat a topic that does not appear in the A-series. They are independent philosophical works, not an appendix to the physics.

Methodological and diagnostic documents (Dok. 241, 242, 244, 247, 248, 250, 262, 277, 281, 284, 287, 288, 303–305 et al.). Hawking information paradox, ratio vs. absolute-value pedagogy, document structure maps, embedding protocols. These documents are working materials that arose during the development of the corpus. They are important for following the development history, but are not part of the canonical state.

Physics extensions in the narrow sense (Dok. 021–024, 027–034, 037–040, 054–068, 075–081, 089, 095, 105, 114, 124, 129, 137, 145–170, 172, 178, 179, 182, 184, 187, 192, 206 et al.). Earlier formulations of the mass formula, alternative parametrisations, intermediate stages of the theory, individual calculations that have been absorbed into A-documents or deliberately not taken over. This block is the largest and most heterogeneous: it contains early correct calculations alongside superseded approaches, independent side-lines alongside already-subsumed substance. The correction register (Dok. 190) books the relevant corrections; the A-documents name what they take over. What appears in neither the register nor an A-document is historical record.

.14 Open Questions

This section appears in every document of the A-series and names what the document does *not* accomplish. The rule is adopted from outside: a completed step that does not name its residual silently claims jurisdiction beyond what it has shown.

For this document, four points remain open.

First, the series does not cover the entire corpus. It orders the geometric core, the unit and constant structure, and the fermion sector; the gauge sector and the dynamic field theory remain in the legacy corpus, as does the early cosmology sector excluded from the series (A240, A250). The commitment to back every computational statement with a verification script has not yet been fulfilled for the series as a whole.

Second, the replacement claim is untested. The blueprint assigns each A-document a set of old numbers whose content it is to subsume. Whether this assignment is complete – whether nothing from the legacy corpus falls through – can only be decided at the finished index A250, not before. Until then, "supersedes" is an intention, not a finding.

Third, the boundary of the cosmological exclusion is not sharp. The exclusion is justified by a difference in standard, not by topic. At three points – the correction factor K_{frak}, the bridge constants, the projection chain – the legacy corpus historically carries cosmological anchors. Whether these points remain valid without such anchors is an open substantive question, not answered by the exclusion decision.

Fourth, the tolerance standard is here only formulated, not applied. The demand to compare every numerical statement against measurement precision and determination tolerance is easy to state. Whether it can be sustained consistently is decided only where numbers appear – that is, from A100 onward.

.15 Supersedes

The correction register referred to is Dok. 190 (entries K1–K6, P1–P43, R1–R56); the winding-number corrections are in Dok. 210.

This document subsumes the framework and overview sections of: Dok. 086 (document overview), Dok. 201 (FFGFT-all, outline section), Dok. 205 (narrative, framework sections). The narrative part of Dok. 205 remains independently worth reading and is not superseded.